

6.6.1 Additional homework ¶

Homework 6.6.1.1. In Units 6.3.1-3 we analyzed how error accumulates when computing a dot product of x and y of size m in the order indicated by

$$\kappa = ((\cdots((\chi_0\psi_0 + \chi_1\psi_1) + \chi_2\psi_2) + \cdots) + \chi_{m-1}\psi_{m-1}).$$

Let's illustrate an alternative way of computing the dot product:

- For $m = 2$:

$$\kappa = \chi_0\psi_0 + \chi_1\psi_1$$

- For $m = 4$:

$$\kappa = (\chi_0\psi_0 + \chi_1\psi_1) + (\chi_2\psi_2 + \chi_3\psi_3)$$

- For $m = 8$:

$$\kappa = ((\chi_0\psi_0 + \chi_1\psi_1) + (\chi_2\psi_2 + \chi_3\psi_3)) + ((\chi_4\psi_4 + \chi_5\psi_5) + (\chi_6\psi_6 + \chi_7\psi_7))$$

and so forth. Analyze how under the SCM error accumulates and state backward stability results. You may assume that m is a power of two.